

Deepak Mallampati

APPLIED AI / FORWARD DEPLOYED ENGINEER

+1-330-554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
Kent, OH (US) - F-1 OPT; open to travel / client embedding

PROFESSIONAL SUMMARY

Applied AI & Forward Deployed Engineer with 4+ years turning messy, real-world operational data into deployed, production-grade tools across energy, enterprise, and regulated domains. Strong Python, SQL, and data-wrangling foundation paired with hands-on delivery of APIs, cloud data pipelines (AWS/Azure/GCP), and ML/LLM systems embedded directly in customer operations. Track record of fast, measurable outcomes - 60% fewer failed pipeline runs, 800ms p95 latency at 100k+ requests/day, and reliable models for equipment-failure prediction and capacity forecasting.

SKILLS

Forward-Deployed Delivery • Messy Data → Production Tools • Python & SQL • Data Ingestion & ETL • Cloud (AWS/Azure/GCP) • APIs & Microservices • ML Modeling & Forecasting • Rapid Prototyping • Client-Facing Delivery • Energy / Utilities Operational Data

Languages & Data: Python, SQL, JavaScript/TypeScript, Java | Pandas, NumPy | PostgreSQL, MongoDB, Redis | Spark, Databricks, Kafka

Cloud & Infrastructure: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure, GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD (Jenkins, Azure DevOps)

ML & Analytics: Scikit-learn, PyTorch, TensorFlow | Regression, Classification, Forecasting | Anomaly Detection | Model Optimization | A/B Testing | RAGAS, BERTScore

Generative AI & LLMs: GPT-4o, Claude, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT)

APIs & Delivery: FastAPI, React | REST APIs & Microservices | Data ingestion patterns | Model monitoring & observability (MLflow, W&B)

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Audit Logging

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer - USA

- Embedded with stakeholders to turn large-scale, messy healthcare document corpora into deployed **retrieval-augmented generation (RAG)** tools, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and reduce hallucination on domain-specific queries.
- Built fault-tolerant **data & automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%**.
- Architected production-grade **agentic LLM systems** on a conductor-subagent framework, decomposing objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy.
- Designed **privacy-preserving data workflows** for sensitive datasets (K-anonymity, L-diversity, differential privacy), enabling compliant analytics with zero PII exposure.
- Established an **evaluation & monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to benchmark iterations and surface regressions before release.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern - USA

- Worked directly with product, platform, and data-engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Developed and enhanced **AI agents and backend services** integrated with production data pipelines and observability tooling across 25+ agents and workflows.
- Engineered safeguards and content controls for policy-compliant responses, reducing unsafe outputs by **42%** while maintaining quality.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant - USA

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset (k-anonymity, l-diversity, Laplace differential privacy across four ϵ levels).
- Reduced record-linkage re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance** (Random Forest).
- Designed a composite **privacy-utility scoring framework** benchmarking six anonymization configurations to identify the optimal operating point.

Emerson

Jun 2022 - Apr 2023

ML Trainee - India

- Developed and trained supervised **regression and classification models** on operational pipeline-storage data for **equipment-failure prediction, anomaly detection, and capacity forecasting**.
- Engineered features and executed end-to-end **data preprocessing** - cleaning, normalization, and transformation of raw sensor and operational datasets - to produce model-ready pipelines.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific datasets, achieving **89% F1-score** on custom NER.

Suvidha Foundation

Jan 2022 - Mar 2022

ML Intern - India

- Designed and built an end-to-end **video summarization and highlight-generation pipeline** (speech-to-text, abstractive summarization, timestamp-aligned clip extraction).
- Deployed across **5,000+ videos**, automating highlight selection and reducing manual editing effort by **70%**.

PROJECTS

career-ops

An autonomous, AI-driven data pipeline that scans listings, scores fit, and generates tailored outputs overnight without supervision - compressing hours of repetitive work into an automated, monitored workflow.

Python · LLM orchestration · scheduled batch automation · retry logic

Privacy-Preserving Clinical ML Pipeline

A framework that lets researchers analyze sensitive patient data without exposing identities, combining k-anonymity, l-diversity, and differential privacy - quantifying exactly how much privacy each technique buys for how little accuracy cost.

Python · Scikit-learn · differential privacy · Random Forest

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - turning a multi-step manual chore into a single inline action.

JavaScript · LLM APIs · real-time inference

Story Director LLM

A fine-tuned model generating structured short-form narratives from a single prompt, paired with an automated rendering pipeline.

QLoRA · Qwen3-27B · NVIDIA H100

RESEARCH

Privacy-Utility Trade-offs in Clinical Machine Learning

Authored a peer-review-ready IEEE-format conference paper documenting methodology, experiments, and results for privacy-preserving ML on clinical data (k-anonymity, l-diversity, differential privacy).

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025

GPA: 3.70 / 4.0